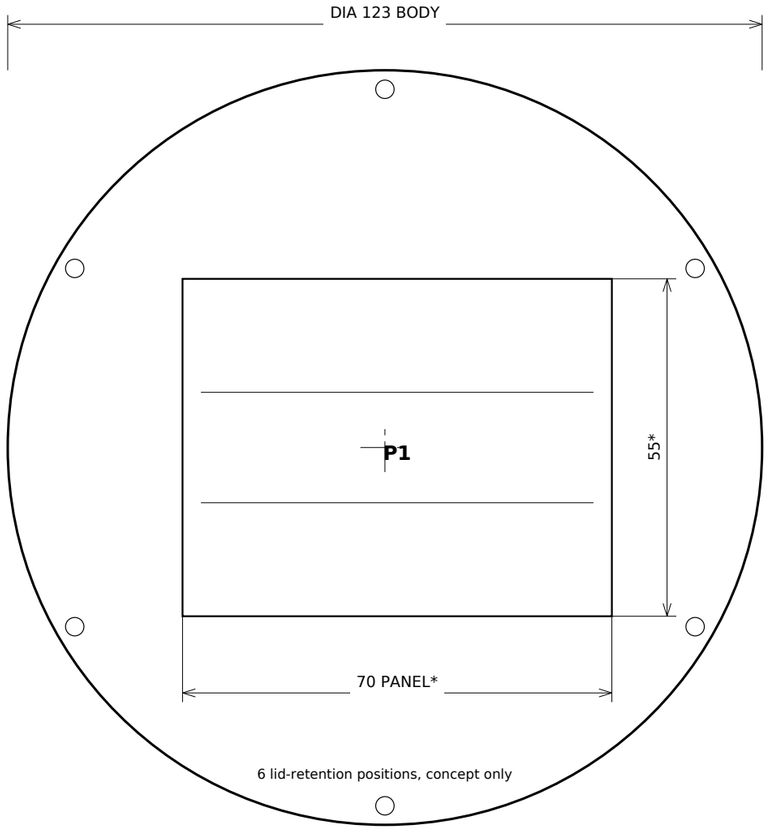
SENSOR PUCK / UPDATED ASSEMBLY

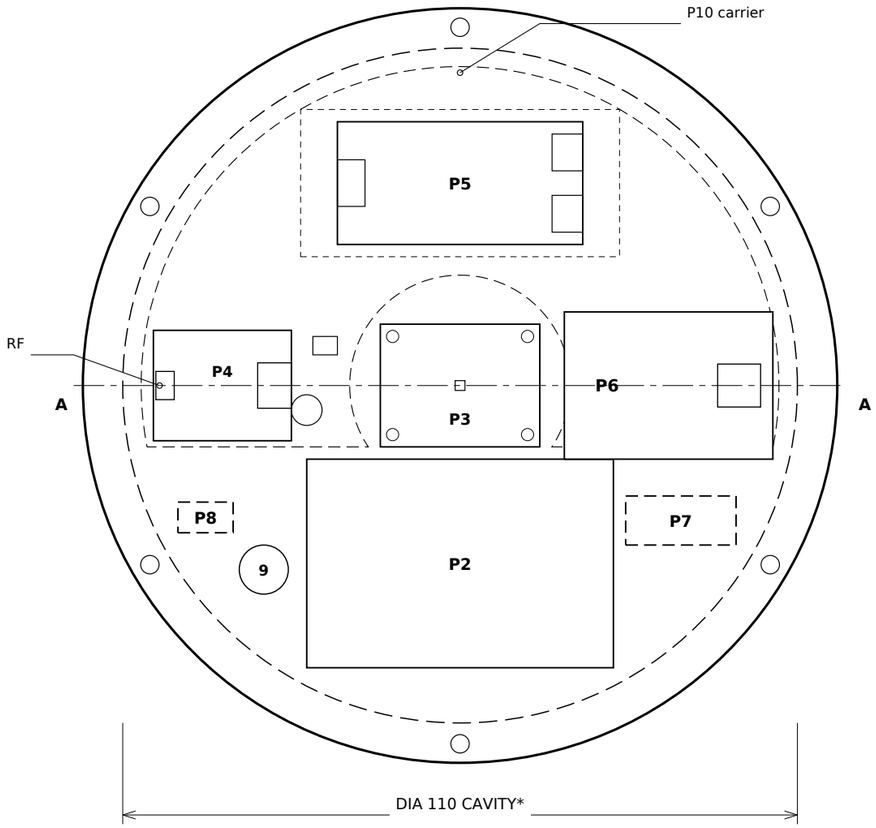
CONCEPT / NOT FOR FABRICATION

True-size orthographic views. Original cylindrical concept retained; internal arrangement revised for the new hardware.

TOP / LID FITTED 1:1



COMPOSITE PLAN / LID REMOVED 1:1



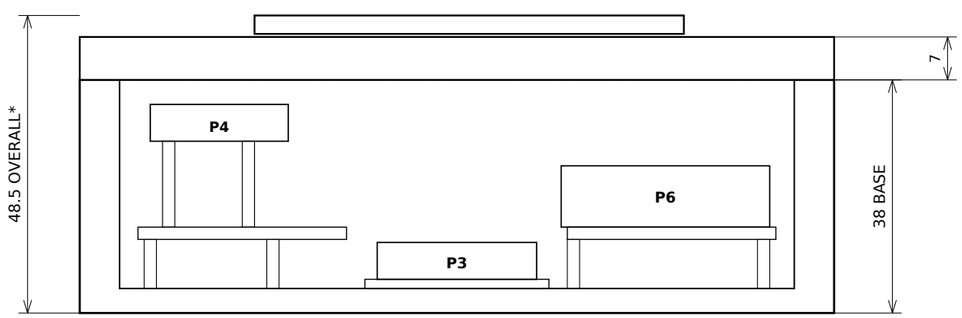
Parts shown at different heights; see section below.

KEY	PART	L x W x T
P1	Panel	70 x 55 x 3*
P2	LiPo	50 x 34 x 8
P3	MMC5603	26 x 20 x 6*
P4	ESP32-C3	22.5 x 18 x 6*
P5	CN3065	40 x 20 x 9*
P6	TPS63020	34 x 24 x 10*
P7	Protection	18 x 8 x 3*
P8	Switch	9 x 5 x 8*
P9	Cap reserve	DIA 8 x 12*
P10	Carrier	2 thick*

* Planning envelope / confirm on receipt.
P3 reference PCB is 25.4 x 17.8;
26 x 20 allows for clone variation.
P7 only if cell protection is inadequate.
P9: optional 220 uF; smaller 10 uF
and 100 nF use local pad space.

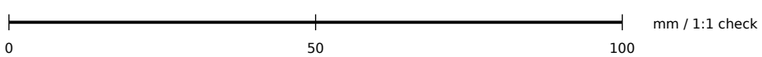
No nylon screw kit assumed.
Use an insulating, removable carrier.
40 x 60 perfboard is cut stock,
not a full board beneath the sensor.

A-A / TRANSVERSE SECTION 1:1



Z = 0 at continuous underside. No floor penetrations.

HEIGHT STACK / mm from underside
0-4 floor; P3 on 1.5 mm sensor sled.
P2 at 8-16; P5 at 14-23; P6 at 14-24.
P4 at 28-34; clear underside of cap at 38.
Cap 38-45; adhesive reserve 0.5; panel 3*.
45 body + 3.5 panel/adhesive = 48.5 overall*.



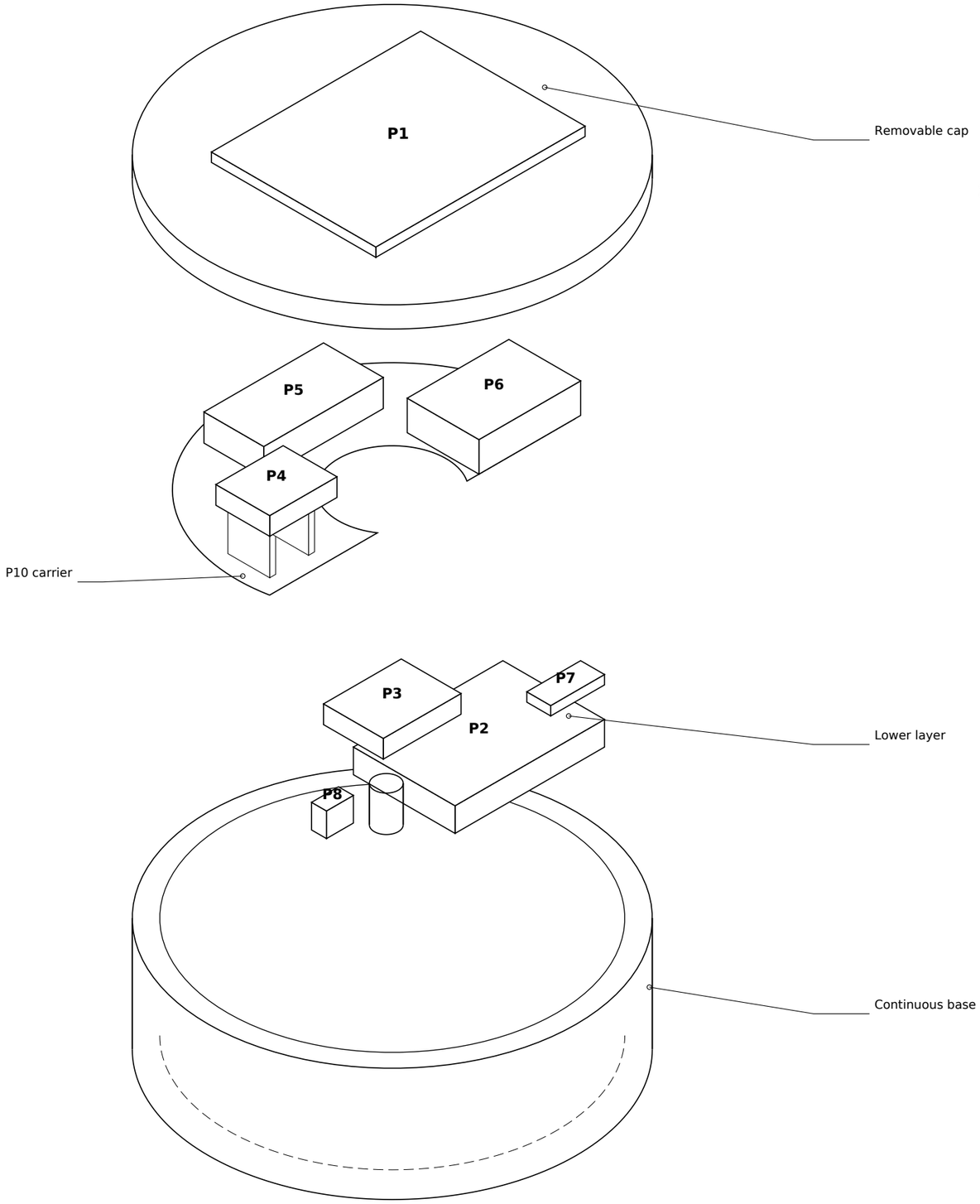
LID + SERVICE
Cap lifts with attached panel.
Disconnect panel lead before removal.
USB access requires opening the lid;
lift MCU/charger if the plug will not seat.
Seals and fasteners remain undetailed.
Steel hardware near P3 needs testing.

FIT STATUS
DIA 110 is an assumed usable cavity.
Minimum nominal radial body margin:
2.6 mm, before leads and tolerances.
Confirm cavity, magnetic baseline and
closed-lid RF before making the carrier.

SENSOR PUCK / RENDERER HANDOFF

Exploded arrangement and enlarged part envelopes. View geometry is editable; no wiring or decorative detail.

EXPLODED AXONOMETRIC / NTS

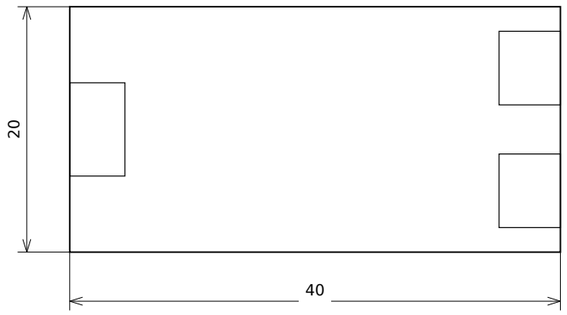


Exploded offsets are illustrative; use sheet 01 for assembled heights.

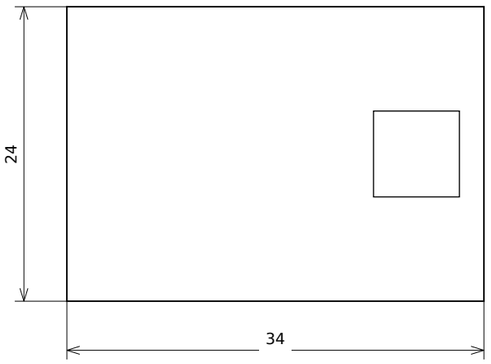
Lid, carrier and sensor sled can be removed independently for showcasing.

PART ENVELOPES / 2:1

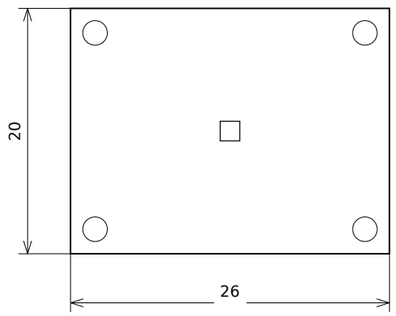
CN3065 / 40 x 20 x 9*



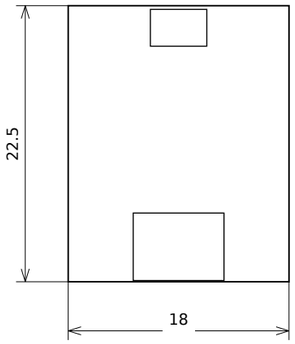
TPS63020 / 34 x 24 x 10*



MMC5603 / 26 x 20 x 6*



SuperMini / 22.5 x 18 x 6*



* Envelopes are not hole patterns. Do not infer drill locations.

MMC5603: reference PCB 25.4 x 17.8 x 4.6; allocation 26 x 20 x 6.

CN3065: reference boards vary about 39-40 x 19-20 x 7-9.

TPS63020: 34 x 24 planning footprint; measure the selected module.

Use low-profile soldered pads. Tall headers/Qwiic plugs are excluded.

Charger has a 6 mm plug reserve at each short end on sheet 01.

No USB cable is stored inside the sealed enclosure.

RENDERING PRIORITIES

Keep the sensor low and central; battery clear of its footprint.

Show the power inductor on the outer/right end of P6.

Do not add steel screws, a full-size perfboard, or internal LoRa.

Panel begins 17 mm inward of the antenna tip in plan; test RF.

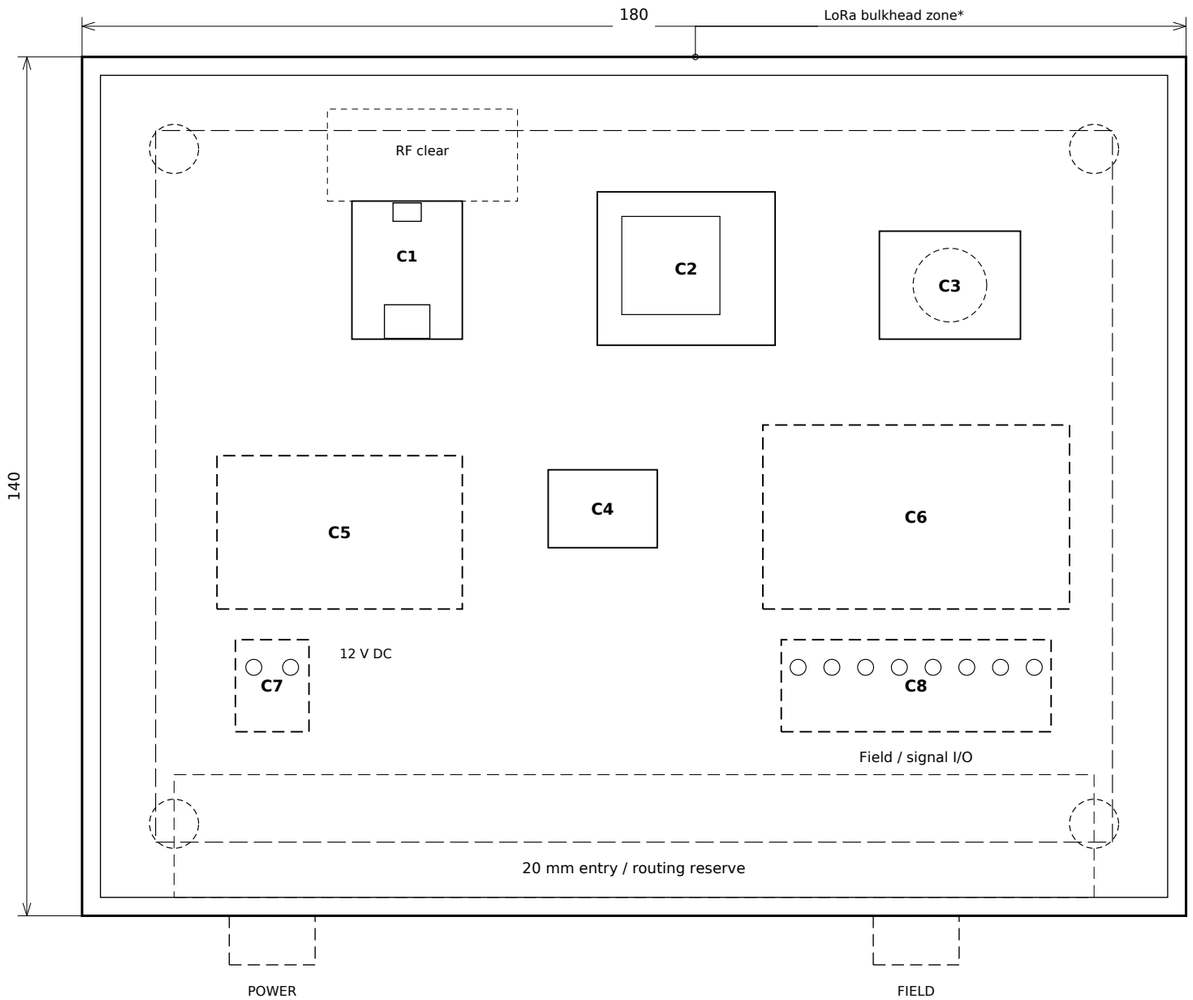
Dimensions/source status and XYZ coordinates are in the pack.

INTERSECTION CONTROLLER / ONE PER CONTROLLER UNIT

CONCEPT / NOT FOR FABRICATION

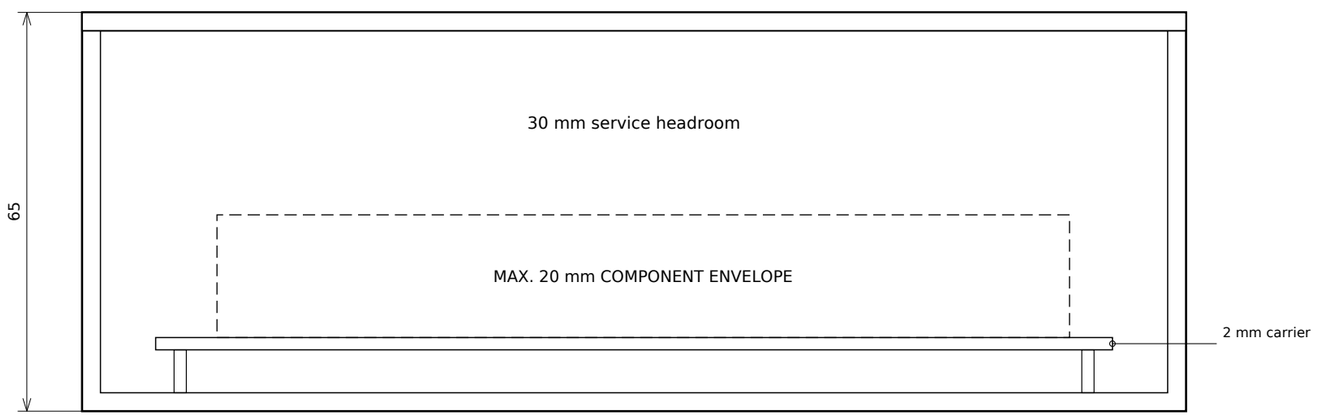
180 x 140 x 65 body, proposed. Each unit has its own LoRa and RTC. Low-voltage supervisory interface assumed.

INTERNAL PLAN / LID REMOVED 1:1



* Port cut-outs and glands require the selected connectors.

DEPTH SECTION / COMPONENT RESERVE 1:1



KEY	REFERENCE / RESERVE	L x W x T
C1	ESP32-C3 SuperMini	18 x 22.5 x 6
C2	LoRa RFM95W breakout	29 x 25 x 8
C3	RTC DS3231 breakout	23 x 17.6 x 9
C4	3.3 V regulator	17.78 x 12.7 x 6
C5	DC protection reserve	40 x 25 x 20
C6	Signal interface reserve	50 x 30 x 20
C7	2-pole power terminal	12 x 15 x 16
C8	Field terminal bank	44 x 15 x 16

C1-C4 use reference module geometry; tall headers excluded.

C5-C8 are reserved volumes, not selected products.

Carrier 156 x 116 x 2; usable cavity 174 x 134 x 59.

3 mm floor + 7 supports + 2 plate + 20 parts + 30 clear + 3 lid = 65.

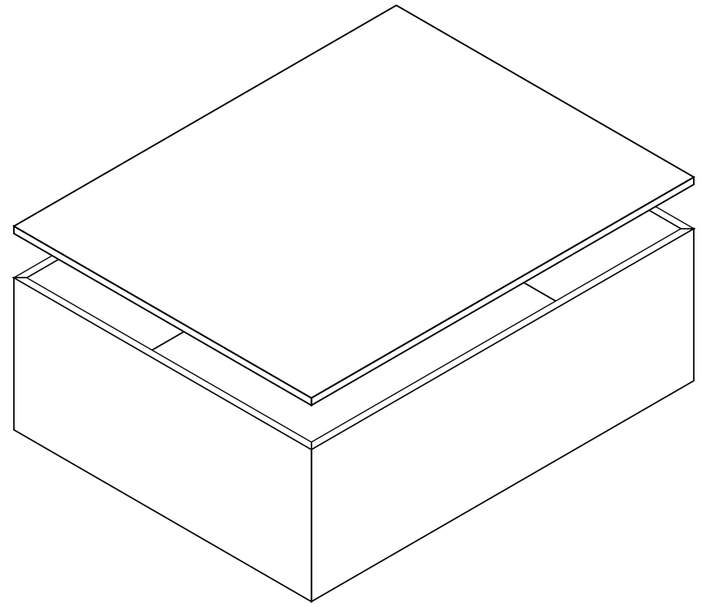
Mount beside the existing signal cabinet

in a nonmetallic sidecar enclosure; gland entries face down.

Keep the ESP32 antenna outside the steel cabinet RF shield.

LoRa antenna, bracket and glands are outside the body size.

LID + BODY / AXONOMETRIC NTS



Supply: external regulated 12 V DC, fused/protected for the site.

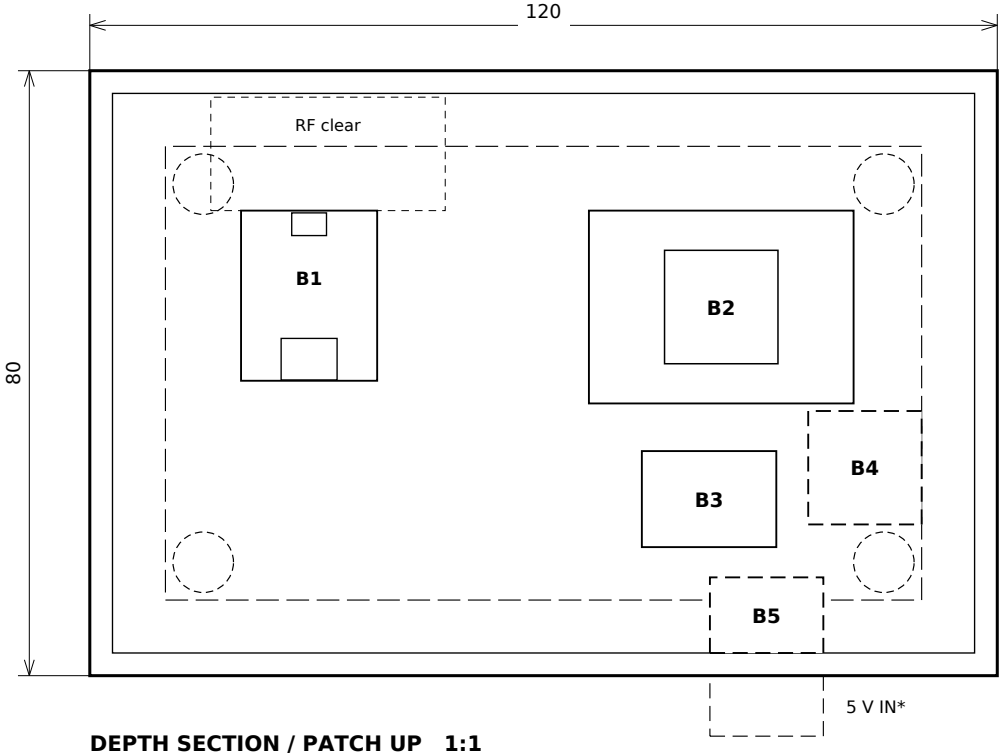
No mains, lamp drivers, UPS or safety interlock hardware included.

Independent LoRa/RTC does not by itself provide controller failover.

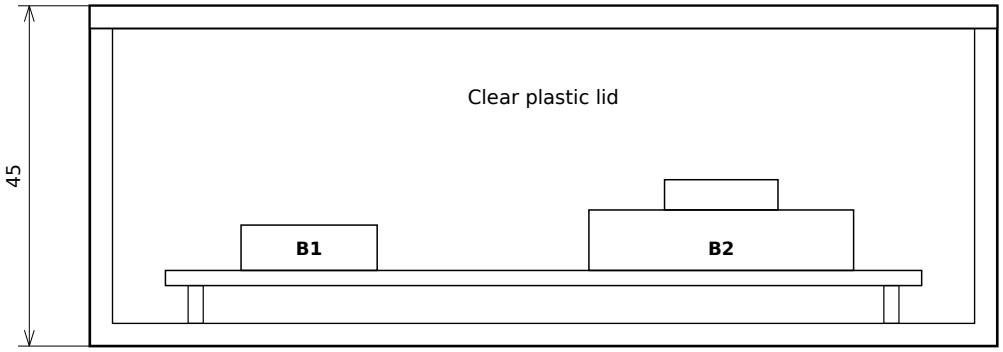
EMERGENCY BEACON / VEHICLE UNIT

120 x 80 x 45 body, proposed. ESP-NOW through plastic lid; GNSS reference module shown.

INTERNAL PLAN / LID REMOVED 1:1



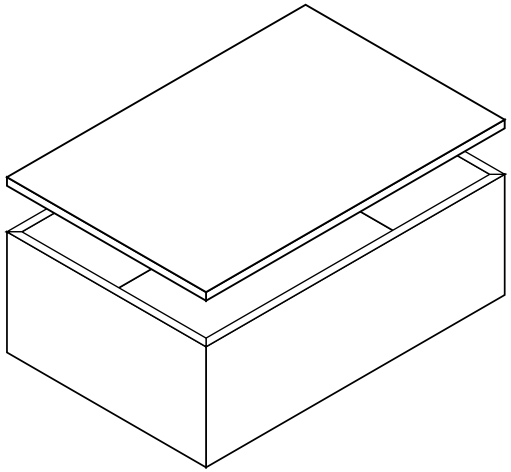
DEPTH SECTION / PATCH UP 1:1



KEY	REFERENCE / RESERVE	X x Y x Z
B1	ESP32-C3 SuperMini	18 x 22.5 x 6
B2	GNSS breakout	35 x 25.5 x 12
B3	3.3 V regulator	17.78 x 12.7 x 6
B4	5 V protection reserve	15 x 15 x 18
B5	5 V connector reserve	15 x 10 x 10

B1-B3: reference geometry. B4-B5: allocated reserves.
GNSS 12 mm envelope includes patch and holder.
Carrier 100 x 60 x 2. Cavity 114 x 74 x 39.
3 floor + 5 posts + 2 carrier + 18 parts
+ 14 headroom + 3 lid = 45 mm.
* Input port and bracket are unselected.

REMOVABLE LID / AXONOMETRIC NTS



Secure near windshield, patch up; verify GNSS reception.
Keep driver view and airbag deployment zones clear.
External protected vehicle adapter: regulated 5 V DC.
No internal battery/LoRa; no raw 12/24 V feed to B5.